SynapseOS: Designing a Conversational Interface Layer for Personal Computing

Formative Assessment 2.1: Chapter 1 Draft

Alexandra Sulit
Willard Soriano
Allyson Vivar

Department of Information Technology
Mapúa University – Makati

CHAPTER 1: INTRODUCTION

(Draft)

1.1 BACKGROUND OF THE STUDY

- The graphical desktop paradigm (icons, menus, pointer-driven interaction) has been the dominant interface for personal computing since the early 1980s
- A parallel research tradition has explored natural language as an alternative interface since the late 1980s, beginning with the Berkeley UNIX Consultant [1]
- LLMs and multimodal foundation models have reopened this research thread with substantially greater technical capability
- Recent progress spans shell-level natural language translation [2, 3], desktop computer-using agents [16, 18], and web agents [8, 9]
- Despite progress, frontier models complete only 12.24% of real computer tasks vs. 72.36% for humans on the OSWorld benchmark [17]
- Coverage is uneven: Windows [15, 16], Android [10, 11, 12], and the browser [7, 8, 9] are heavily represented; Linux desktop is conspicuously absent
- No existing system implements a full-OS conversational replacement for Linux — only automation on top of conventional desktops

1.2 GAP / OPPORTUNITY

- No full-OS conversational replacement exists for the Linux desktop, despite Linux being the substrate for most developer and server-side workflows [3, 17]
- Existing systems automate on top of the conventional desktop rather than replacing it [16, 18]
- No prior work compares a conversational interface against the conventional graphical workflow on the same tasks with the same users [17]
- Human-centered evaluation of LLM-based interaction is rare; only VoicePilot [5] conducts a substantive user study in this space

1.3 OBJECTIVES / PROBLEM STATEMENT

General Objective

- Design, implement, and evaluate SynapseOS, a Linux-based conversational interface layer intended to replace the conventional graphical desktop as the primary user-facing interface

Specific Objectives

- Design a conversational interface layer for the Linux desktop spanning shell operations, graphical applications, and cross-application workflows
- Develop a hybrid execution architecture: MCP-exposed APIs → AT-SPI accessibility tree → vision-based GUI simulation fallback
- Implement SynapseOS as a Linux session manager with persistent conversational surface, undo log, and confirmation policy for irreversible operations [3, 16]
- Evaluate SynapseOS against the conventional Linux desktop on task completion time, error rate, and user satisfaction across novice and power user populations [17]

REFERENCES

[1] Wilensky, R., Chin, D. N., Luria, M., Martin, J., Mayfield, J., & Wu, D. (1988). The Berkeley Unix Consultant Project. *Computational Linguistics*, 14(4), 35–84. https://aclanthology.org/J88-4003/

[2] Westenfelder, F., Hemberg, E., Tulla, M., Moskal, S., O'Reilly, U.-M., & Chiricescu, S. (2025). LLM-Supported Natural Language to Bash Translation. *arXiv:2502.06858*. https://arxiv.org/abs/2502.06858

[3] Gyawali, B. R., Achalla, S., Kallas, K., & Kumar, S. (2025). NaSh: Guardrails for an LLM-Powered Natural Language Shell. *arXiv:2506.13028*. https://arxiv.org/abs/2506.13028

[5] Padmanabha, A., Yuan, J., Gupta, J., Karachiwalla, Z., Majidi, C., Admoni, H., & Erickson, Z. (2024). VoicePilot: Harnessing LLMs as Speech Interfaces for Physically Assistive Robots. In *Proceedings of UIST '24*. https://doi.org/10.1145/3654777.3676401

[7] Deng, X., Gu, Y., Zheng, B., Chen, S., Stevens, S., Wang, B., Sun, H., & Su, Y. (2023). Mind2Web: Towards a Generalist Agent for the Web. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. https://arxiv.org/abs/2306.06070

[8] Zhou, S., Xu, F. F., Zhu, H., Zhou, X., Lo, R., Sridhar, A., Cheng, X., Ou, T., Bisk, Y., Fried, D., Alon, U., & Neubig, G. (2024). WebArena: A Realistic Web Environment for Building Autonomous Agents. In *Proceedings of ICLR 2024*. https://arxiv.org/abs/2307.13854

[9] Zheng, B., Gou, B., Kil, J., Sun, H., & Su, Y. (2024). GPT-4V(ision) is a Generalist Web Agent, if Grounded. In *Proceedings of ICML 2024*. https://arxiv.org/abs/2401.01614

[10] Rawles, C., Li, A., Rodriguez, D., Riva, O., & Lillicrap, T. (2023). Android in the Wild: A Large-Scale Dataset for Android Device Control. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. https://arxiv.org/abs/2307.10088

[11] Zhang, C., Yang, Z., Liu, J., Han, Y., Chen, X., Huang, Z., Fu, B., & Yu, G. (2023). AppAgent: Multimodal Agents as Smartphone Users. *arXiv:2312.13771*. https://arxiv.org/abs/2312.13771

[12] Wang, J., Xu, H., Ye, J., Yan, M., Shen, W., Zhang, J., Huang, F., & Sang, J. (2024). Mobile-Agent: Autonomous Multi-Modal Mobile Device Agent with Visual Perception. *arXiv:2401.16158*. https://arxiv.org/abs/2401.16158

[15] Hui, T., Zhang, Y., Jiang, B., Sun, J., Cheng, F., Lin, X., & Yang, Y. (2025). WinClick: GUI Grounding with Multimodal Large Language Models. *arXiv:2503.04730*. https://arxiv.org/abs/2503.04730

[16] Zhang, C., He, S., Qian, J., Li, B., Li, L., Qin, S., Kang, Y., Ma, M., Liu, G., Lin, Q., Rajmohan, S., Zhang, D., & Zhang, Q. (2025). UFO²: The Desktop AgentOS. *arXiv:2504.14603*. https://arxiv.org/abs/2504.14603

[17] Xie, T., Zhang, D., Chen, J., Li, X., Zhao, S., Cao, R., Hua, T. J., Cheng, Z., Shin, D., Lei, F., Liu, Y., Xu, Y., Zhou, S., Savarese, S., Xiong, C., Zhong, V., & Yu, T. (2024). OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. https://arxiv.org/abs/2404.07972

[18] Zhang, C., He, S., Qian, J., Li, B., Li, L., Qin, S., Kang, Y., Ma, M., Liu, G., Lin, Q., Rajmohan, S., Zhang, D., & Zhang, Q. (2024). Large Language Model-Brained GUI Agents: A Survey. *arXiv:2411.18279*. https://arxiv.org/abs/2411.18279